

CELLCHECK · DESIGN DECISIONS REGISTER

CellCheck design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Estimated cost updated to \$177.60 (\$2.60 over the target) after the fan finger guards and silicone were priced; open decision 1 added for the overshoot
0.4	2026-10-02	Amish Chadha	Design for construction (CCK-DDR-003, P1 to P12) accepted by Amish on 2026-10-02; the 2026-10-01 row no longer says open for review
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations of open items 1 to 11 on 2026-10-02; all moved to decisions made; value-engineering savings line updated
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note and decision records, the build plan work and every decision made so far

CellCheck design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Open decisions.

#	Decision	Options	Recommendation
1	The estimated cost is \$2.60 over the \$175 value-engineering target once the fan finger guards (line 10, \$1.20) and the silicone (line 16, \$4.00) decided on 2026-10-02 are priced	(a) Accept the overshoot: the target is a control target, not a limit. (b) Look for savings of \$2.60 or more, for example a cheaper holder or charger module (each saving is multiplied by eight). (c) Raise the target, which <code>budget_usd</code> would follow	(a) for the paper design; revisit with real quotes at TRL 4 purchasing. Proposed, awaiting Amish

■ To confirm when parts are bought

Table 2. Items to check against the parts before building.

#	What to confirm	Why it matters	Source
1	The steel tray is plain steel, about 460 x 300 x 45 mm with 1.5 mm walls, and its flat floor is at least 457 x 297 mm inside	The liner, plate and containment estimate assume it	CCK-DDR-003, P1
2	The rubber feet have M4 studs about 8 mm long	The stud must pass the 1.5 mm floor and engage the standoff	CCK-DDR-003, P1
3	The M4 rivet nuts grip 1.5 mm sheet and their heads are under 1 mm thick	The guard flange sits round the head	CCK-DDR-003, P4
4	The cell holders have two M3 fixing holes in their base and take 18650 and 21700 cells up to 70 mm	Holder screws and the 84 mm holder length	CCK-DDR-003, P3
5	The charger modules, current monitor breakouts, controller and converter have M3 mounting holes (or are fixed to a small perfboard that has them)	Each board stands on two nylon standoffs	CCK-DDR-003, P5
6	The 2.8 in display module is about 86 x 50 mm with four corner holes	Display stand size	CCK-DDR-003, P6
7	The heatsink spine is 6 mm thick, 50 mm tall, and has a fin gap at 20 mm from the left end and every 40 mm after it	MOSFET screws go between fins; plate and shroud screws are tapped into the spine	CCK-DDR-003, P7
8	The fans have a 50 mm square screw pattern	Shroud holes	CCK-DDR-003, P8
9	The DC jack, fuse holder and rocker switch cut-out sizes against the inlet housing (11.2 mm, 12 mm, 12 x 16 mm)	Inlet housing holes	CCK-DDR-003, P10
10	A sourced figure for the heat released by one 21700 cell in thermal runaway, to replace the assumed 100 kJ	Containment estimate before any TRL 4 work	CCK-CAL-001 section 7

■ Value engineering

Value-engineering target: USD 175.00 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 177.60 (USD 2.60 over the target).

Main cost drivers (CCK-CAL-001, section 11): the per-channel parts, which are bought eight times (cell holder with four-wire contacts, charger module, channel board, load MOSFET and op-amp loop, thermistor: about \$66 for the eight channels), the mains adapter (\$15), the steel tray with liner (\$12) and the controller board (\$10). The parts added for construction (lines 19 to 21) cost \$8.40.

Savings worth trying, should indicative prices rise above the target:

- Leave the matched-group bins out of the BOM until TRL 4 purchasing (decided 2026-10-02). The fan finger guards (line 10, \$1.20 for two) and the silicone for the wire slots (line 16, \$4.00) decided on 2026-10-02 are now in the estimate; they are the reason it is over the target (open decision 1).
- Look for a cheaper cell holder or charger module, since each saving is multiplied by eight.

■ Decisions made

Table 3. Decisions made, with dates and records.

Date	Decision	Decided by	Record
2026-09-25	Keep SwapCell in the pitch and plan a second-life variant; charge and discharge attended only (rest stage may be left); hardware watchdog and undervoltage comparators; grading thresholds, 14-day rest and ± 1 % group tolerance as starting values; TP5100, INA226 and ESP32-S3 class modules; eight channels with a linear load; LFP profile later; per-cell CSV mapped to the ReflowEconomy passport (items 1 to 3 and 5 to 9)	Amish: "i accept all your recommendations, go with them across all repos."	CCK-DDR-001, CCK-DDR-002
2026-09-25	Budget raised from \$160 to \$175 (item 4)	Amish, same instruction	CCK-DDR-001, CCK-DDR-002
2026-09-25	DC four-wire resistance and a 14-day rest (item 11); firmware and build rules: same-channel cohort self-discharge reading, temperature-gated resistance pulse, per-channel calibration, fan-fault stop, detached-thermistor check (item 13); raise the passport gaps with ReflowEconomy (item 15)	Amish, same instruction	CCK-DDR-001, CCK-DDR-002
2026-10-01	Design for construction: feet and standoffs, 1.5 mm plate on six standoffs, holder fixings and wire slots, folded guard with thumb screws, board standoffs, larger display stand, heatsink and shroud fixings, thermistor C-clips, inlet housing, 5 V converter and channel boards (P1 to P12)	Made under Amish's 2026-09-30 instruction: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The changes themselves were accepted on 2026-10-02 (below)	CCK-DDR-003
2026-10-02	Open item 1: the guard is held down by the two knurled M4 thumb screws for the first prototype, and the TRL 4 containment test is run with the guard held exactly that way; latches are revisited only if the test or users show the screws being left loose	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-003, A1
2026-10-02	Open item 2: each wire slot under the guard is filled round its leads with high-temperature silicone after wiring, closing the guard space again	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-003, A2
2026-10-02	Open item 3: first trial partner, the first candidate to approach (not yet agreed), is a local Repair Café group that already passes laptop batteries to an e-waste or battery collection point, with that collection point as the source of trial cells	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 10
2026-10-02	Open item 4: scheduling rule for the attended day: a charge or discharge starts only if it will finish the same day (12.0 cells per day); no overnight pause	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 12
2026-10-02	Open item 5: R11, the 4.91 kg mass is accepted with the 1.5 mm tray kept, and the prototype is weighed at TRL 4; R5, throughput rests on the	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 14; CCK-CAL-001 section 8

Date	Decision	Decided by	Record
	same-day rule of item 4, and more channels are left for a later version		
2026-10-02	Open item 6: 60 mΩ stays the grade A default; per-model limits are added only for the few cell models most common in the partner's intake, set from DC resistance measured on known-good cells of each model, not from data sheet impedance	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 16
2026-10-02	Open item 7: 18650 cells accepted in the product renders; the model keeps the 21700 envelope for clearance checks	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 8: the matched-group bins are render context only for now; a BOM line is decided at TRL 4 purchasing	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 9: fan finger guards are added to BOM line 10 now, with an estimated price, rather than at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 10: the guard's slotted perforation and 1.0 mm sheet in the renders are accepted as appearance only; the BOM keeps round holes in 0.8 mm sheet	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 11: the second-life pack study (about 334 Wh, 165 W) is sent to the SwapCell and PowerBox projects as a proposal; no further CellCheck action	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 1; CCK-CAL-001 section 10
2026-10-02	Design for construction accepted: the changes P1 to P12 and their knock-on changes, as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	CCK-DDR-003, Tables 1 and 2